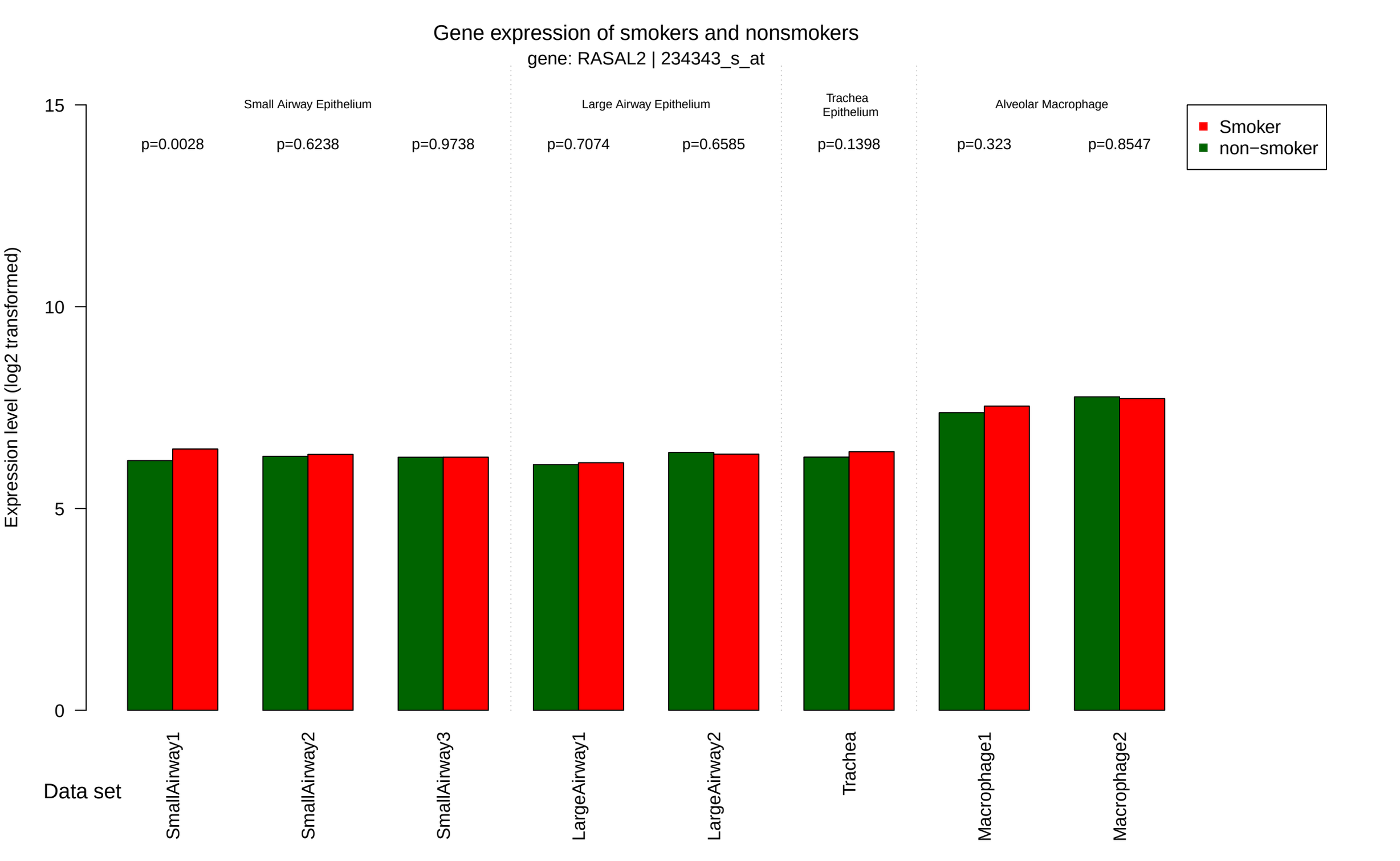
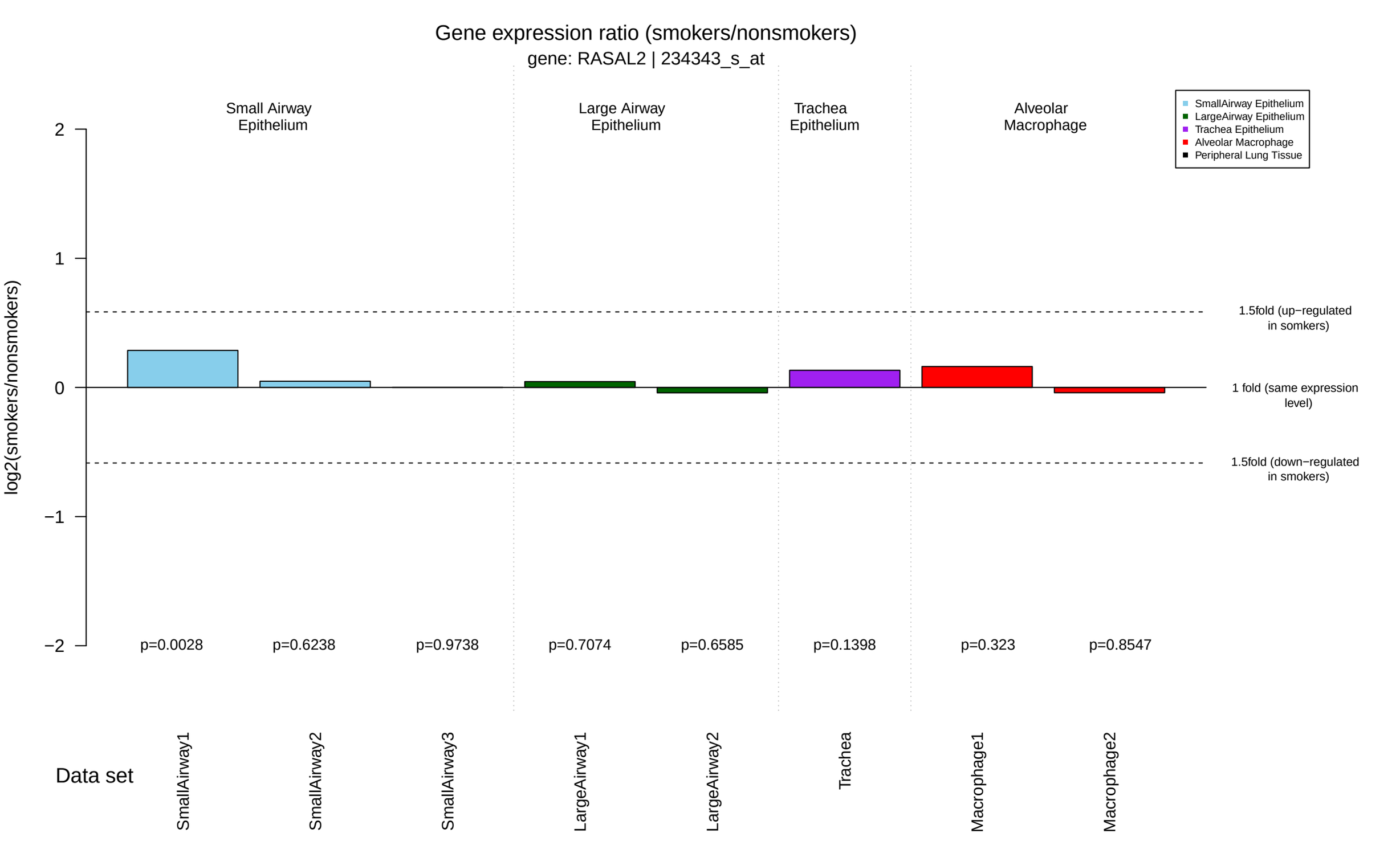
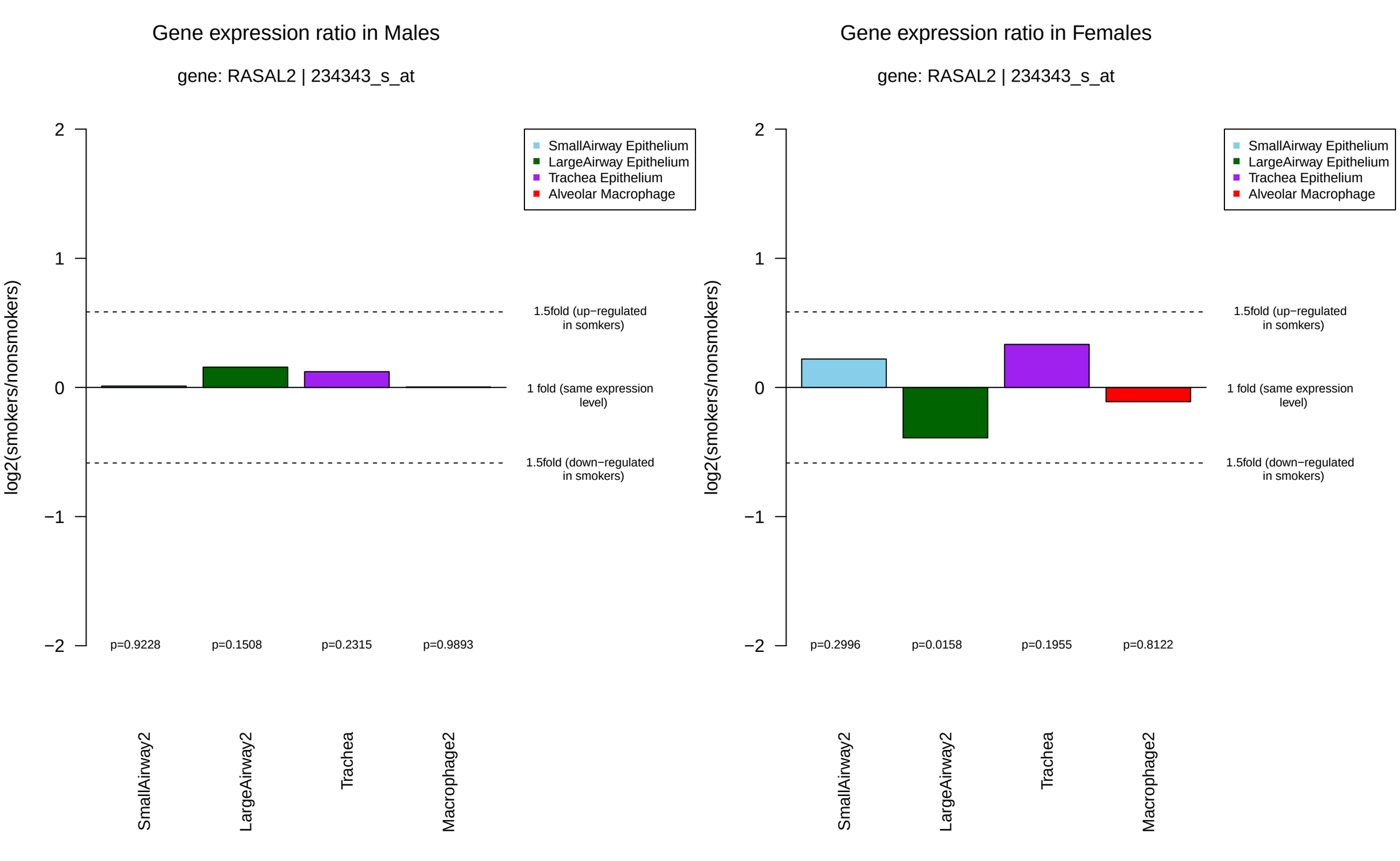
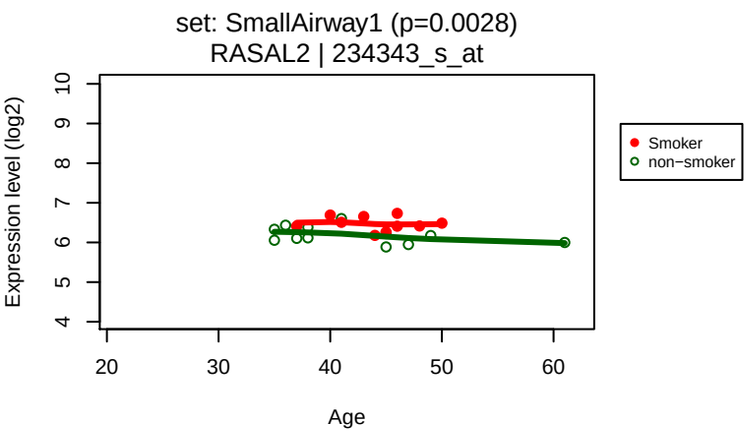
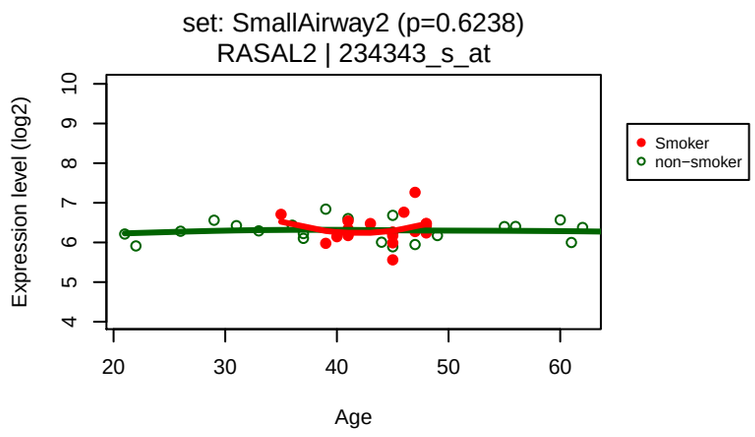
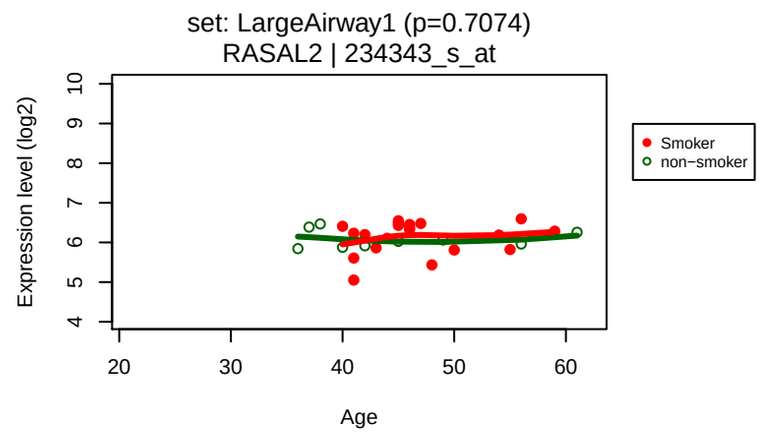

Gene expression of smokers and nonsmokers
gene: RASAL2 | 234343_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0028
p=0.6238
p=0.9738
p=0.7074
p=0.6585
p=0.1398
p=0.323
p=0.8547
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

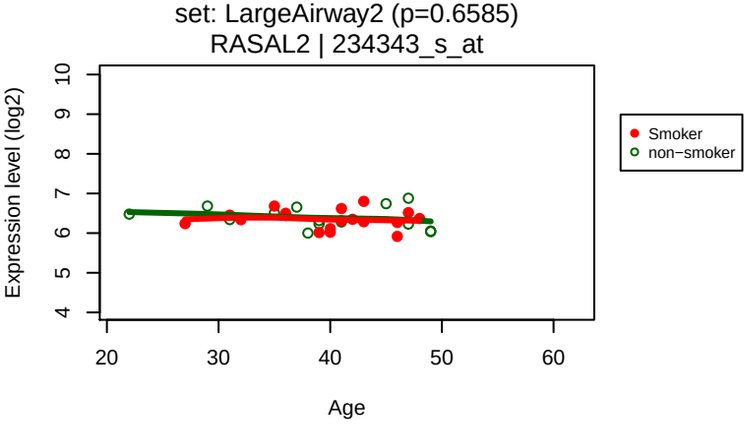

Gene expression ratio (smokers/nonsmokers)
gene: RASAL2 | 234343_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0028
p=0.6238
p=0.9738
p=0.7074
p=0.6585
p=0.1398
p=0.323
p=0.8547
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

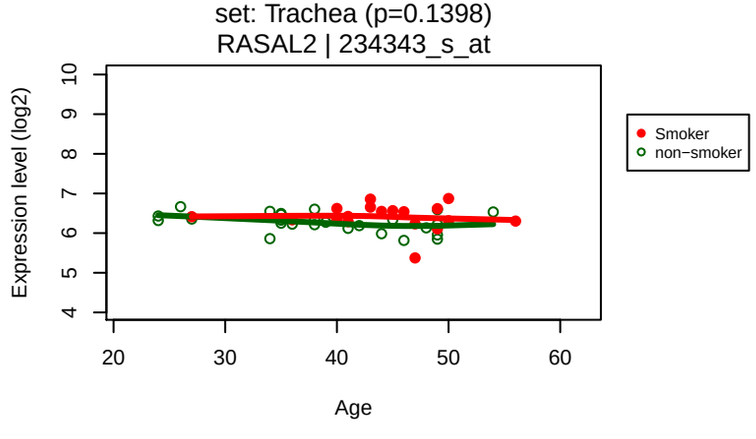

Gene expression ratio in Males
gene: RASAL2 | 234343_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9228
p=0.1508
p=0.2315
p=0.9893
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASAL2 | 234343_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2996
p=0.0158
p=0.1955
p=0.8122
SmallAirway2
LargeAirway2
Trachea
Macrophage2

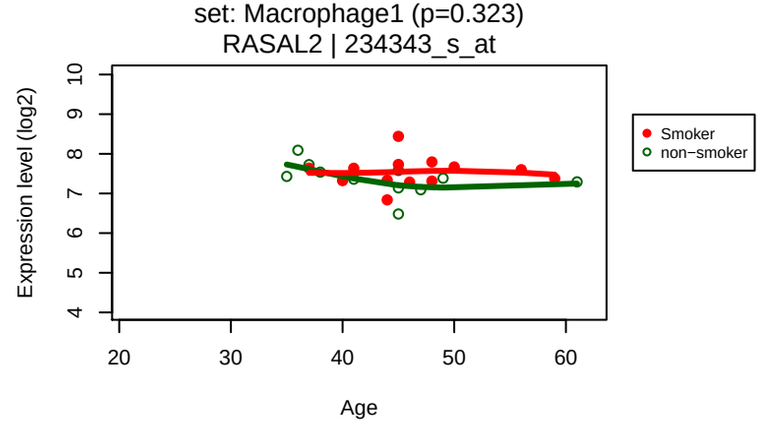

set: SmallAirway1 (p=0.0028)
RASAL2 | 234343_s_at
Expression level (log2)
Age
Smoker
non−smoker

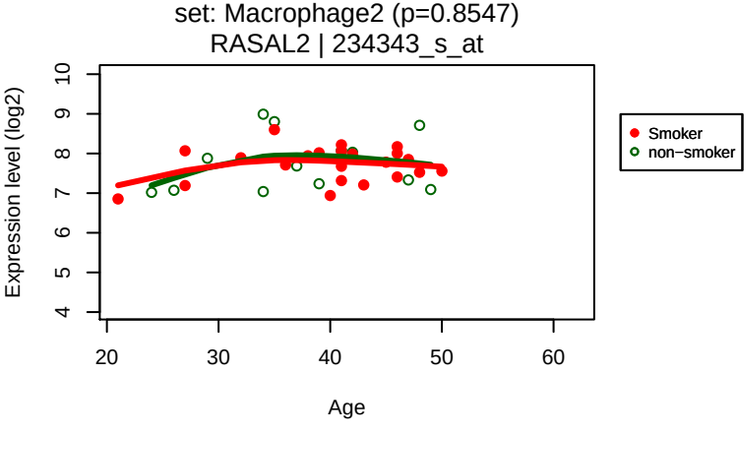

set: SmallAirway2 (p=0.6238)
RASAL2 | 234343_s_at
Expression level (log2)
Age
Smoker
non−smoker

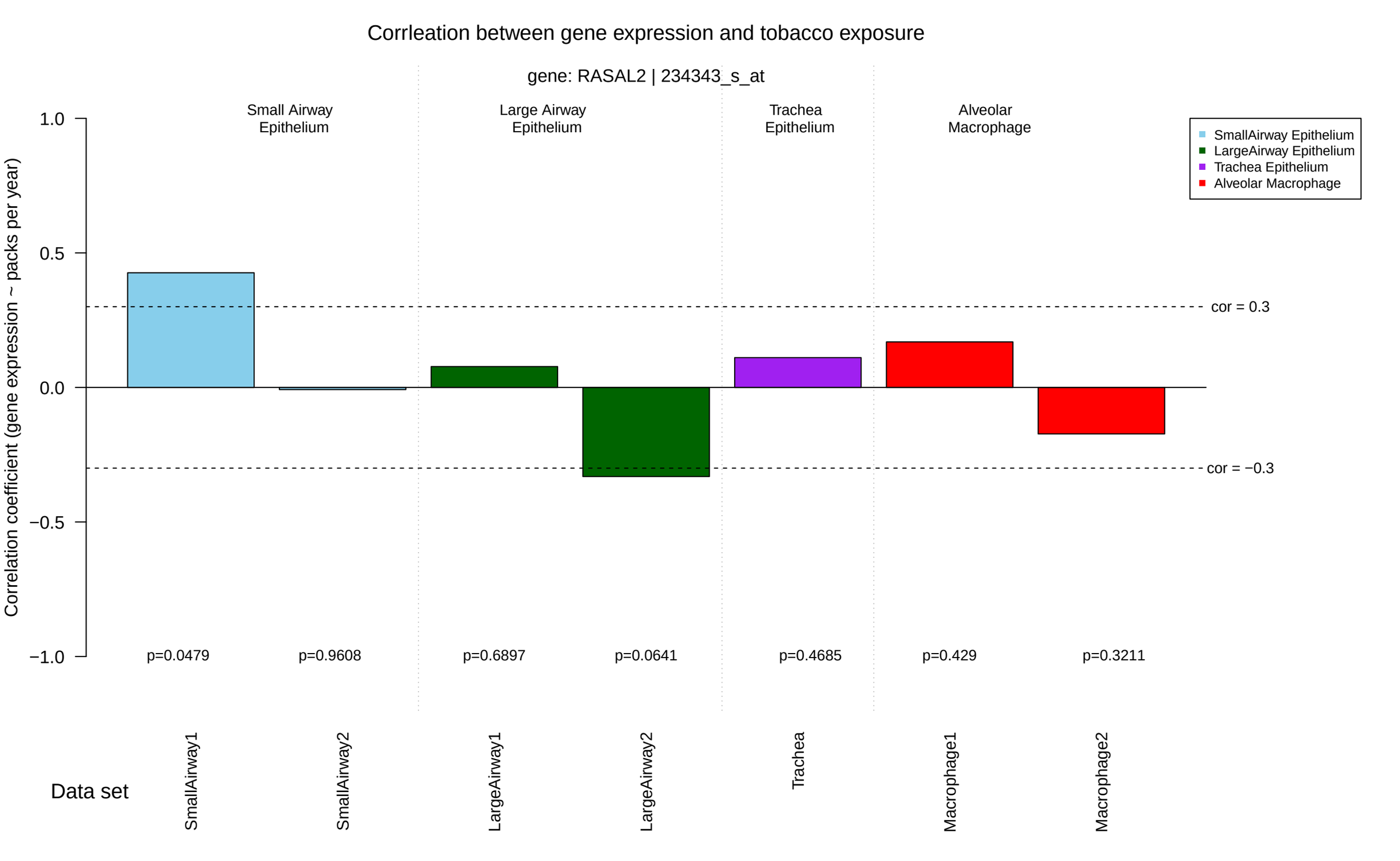

set: LargeAirway1 (p=0.7074)
RASAL2 | 234343_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6585)
RASAL2 | 234343_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1398)
RASAL2 | 234343_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.323)
RASAL2 | 234343_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8547)
RASAL2 | 234343_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASAL2 | 234343_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0479
p=0.9608
p=0.6897
p=0.0641
p=0.4685
p=0.429
p=0.3211
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2